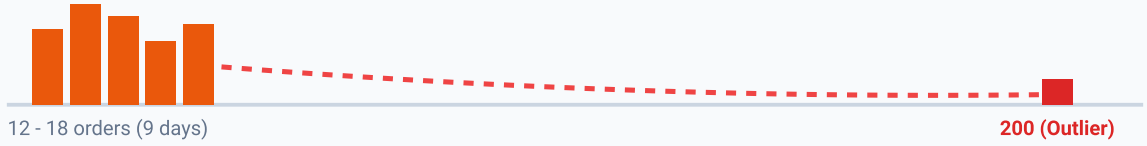
Session 4 Skewness(Basic)

Understanding Asymmetry, Outlier Tails & Data Distributions

1. Swiggy Daily Order Counts (Distribution & Skewness)

Dataset: 10-day order counts = `[12, 15, 17, 14, 13, 16, 200, 18, 14, 15]`

ORDER COUNT DISTRIBUTION PLOT



CLASSIFICATION & REASONING

Distribution Type: Right-Skewed (Positively Skewed)

Reasoning: 9 out of 10 days show consistent order counts tightly clustered between 12 and 18. However, day 7 has an extreme outlier (200 orders), creating a long tail extending far to the right end of the graph and pulling the arithmetic mean (32.4) significantly higher than the median (15.0).

2. Instagram Influencer Follower Counts

Dataset (in thousands): `[5, 7, 8, 8, 9, 10, 12, 15, 95]` (N = 9)

Calculation Breakdown

- **Sum:** 5 + 7 + 8 + 8 + 9 + 10 + 12 + 15 + 95 = 169
- **Mean (μ):** 169 / 9 = **18.78k**
- **Sorted Midpoint (5th value): 9.00k**

Comparison & Metric

Mean (18.78k) > Median (9.00k)

The mean is over twice as large as the median value.

SKEWNESS CONCLUSION

Since the **mean is pulled strongly to the right of the median** (18.78 > 9), the dataset is **Right-Skewed (Positively Skewed)**. The single massive influencer account with 95,000 followers skews the average upward, making the median a much more representative measure of typical follower count.

3. Real-World Right-Skewed Example: YouTube Video Views

Dataset: Public Video View Counts on a Content Creator's YouTube Channel

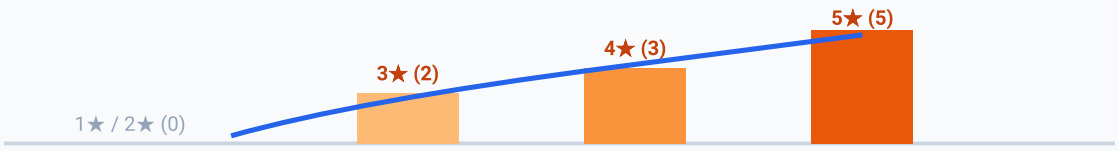
Description: A channel typically has dozens of regular uploads that garner modest viewership (e.g., between 1,000 and 5,000 views per video), alongside 1 or 2 viral videos that hit 2,000,000+ views.

Why it is Right-Skewed: View counts have a strict lower bound at zero (a video cannot receive negative views), constraining the left side. Most videos cluster near the lower-to-middle range, while a tiny percentage of viral hits create a massive, long tail extending to the right.

4. Flipkart Product Ratings Distribution

Dataset: 10 product ratings = `[3, 3, 4, 4, 4, 5, 5, 5, 5, 5]`

RATING FREQUENCY HISTOGRAM



MEAN RATING
4.30 / 5.0

MEDIAN RATING
4.50 / 5.0

ONE-LINE JUSTIFICATION

The distribution is Left-Skewed (Negatively Skewed) because most ratings cluster at the upper maximum (5 stars) while a elongated tail of lower ratings extends toward the left, pulling the mean (4.30) below the median (4.50).